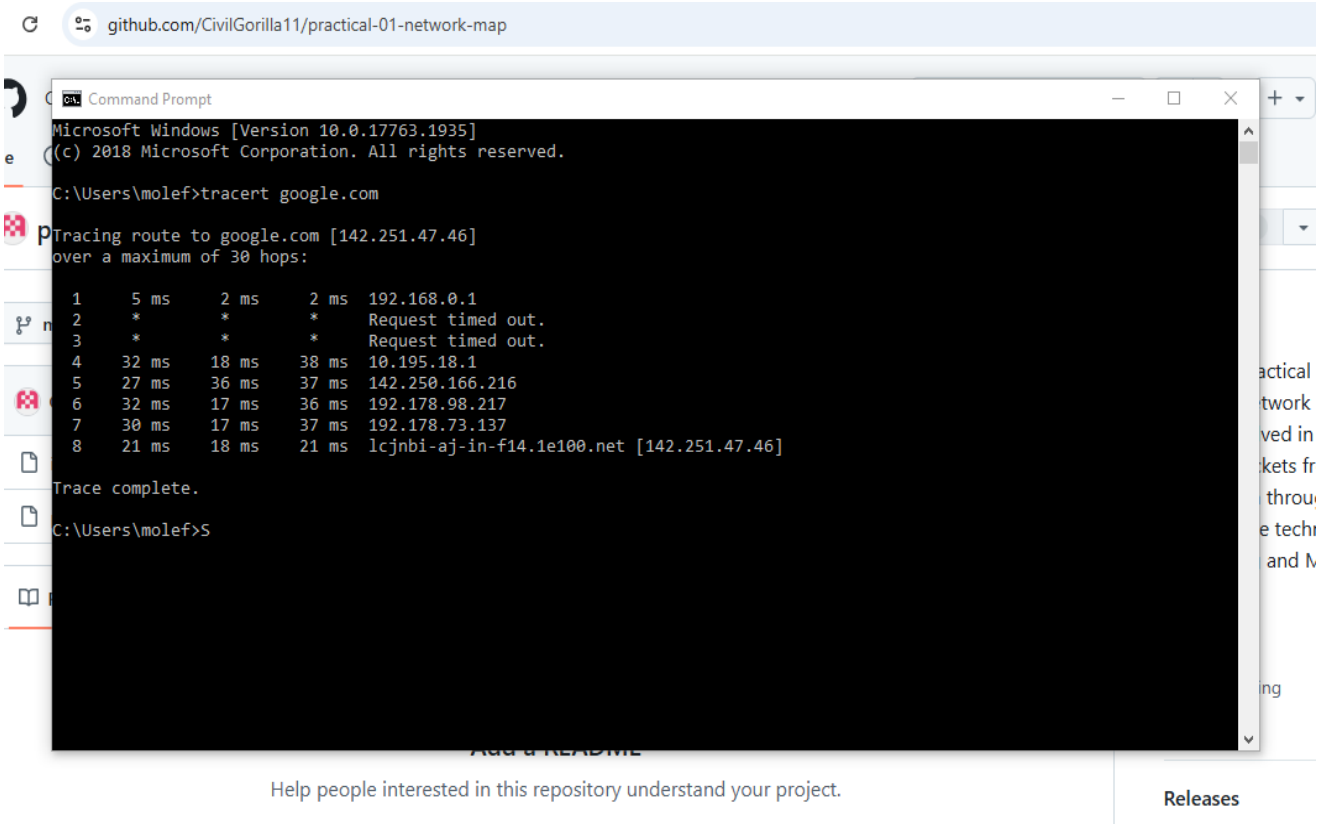

We have ran a traceroute command in our Windows cmd to the web server google.com.

- Hops represent the number of routers that your router will go through to reach the destination server. In our screenshot above we can see that our machine has taken 8 Hops to reach the google.com web server.
- The Round Trip Time is the time measured from when my machine sends the tracert prompt out to a router to when it receives the reply back. The RTT in my first hop is 5ms and the RTT at my final hop is 21ms. The more the RTT means the longer the distance between router and host and server and the router.
- We can see several hops show *** asterick at hop 2 and hop 3. These astericks represent a request time out. This can happen because;
    - Router firewall was configured to silently drop the connection
    - Security policy to reduce attack surface

The only time that the *** are a problem is if they appear in our final hop. That would mean that the there is no router to forward the tracert to. Normally when *** appear in the middle of our hops they are still forwarded to the next hop even though the current hop has silently dropped the request.

- For our tracert command in my machine we can also identify that the hops all have different IP Addresses. These IP Addresses help us identify which hops fall within my ISP and which fall outside of the ISP range.
  Hop 4 represents the ISP IP with a RTT of 38ms
  Hop 5 Represents an IP outside our private home range with RTT of 37ms
  Hop 8 Also represent an IP outside our home range with an RTT of 21ms